1. Question 1: Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

- Cepstral Features

  The cepstral heatmap shows that lower-order coefficients dominate, capturing the spectral envelope of speech. Smooth temporal transitions indicate proper framing and windowing. Higher-order coefficients remain near zero, reflecting noise suppression and dimensionality reduction.

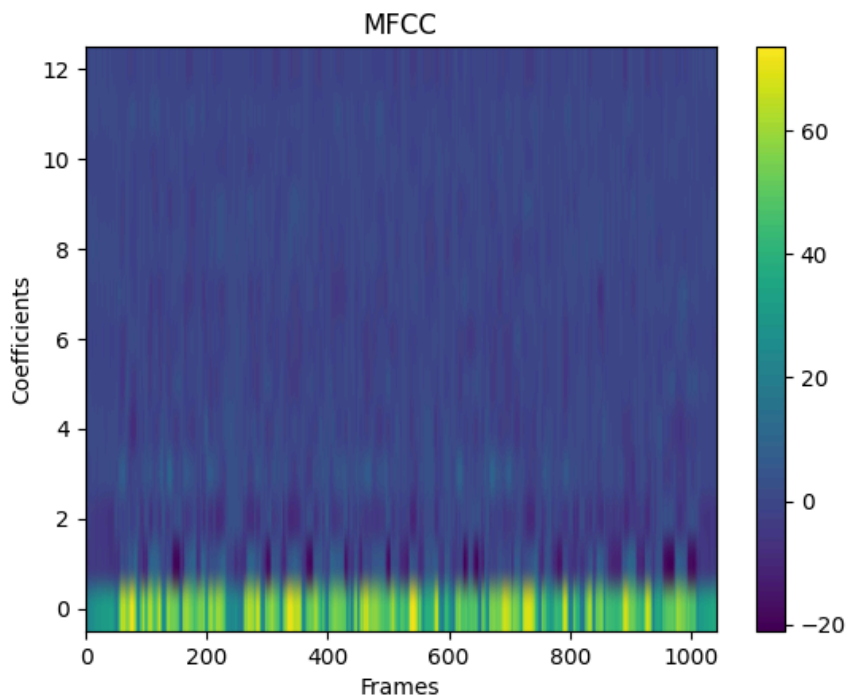

  Figure 1: Cepstral features heatmap across time frames. The x-axis represents frame indices, while the y-axis represents cepstral coefficients. Color intensity indicates coefficient magnitude.

- Spectral Leakage

  Visual inspection:

  - The rectangular window shows highest spectral leakage due to abrupt discontinuities.
  - Hanning window provides best leakage suppression.

- Hamming window balances resolution and leakage
- Energy concentration is highest near main lobe for Hanning/Hamming

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rectangular -> Leakage: 0.9725, SNR: inf dB
hamming -> Leakage: 0.9722, SNR: 4.63 dB
hanning -> Leakage: 0.9732, SNR: 3.90 dB
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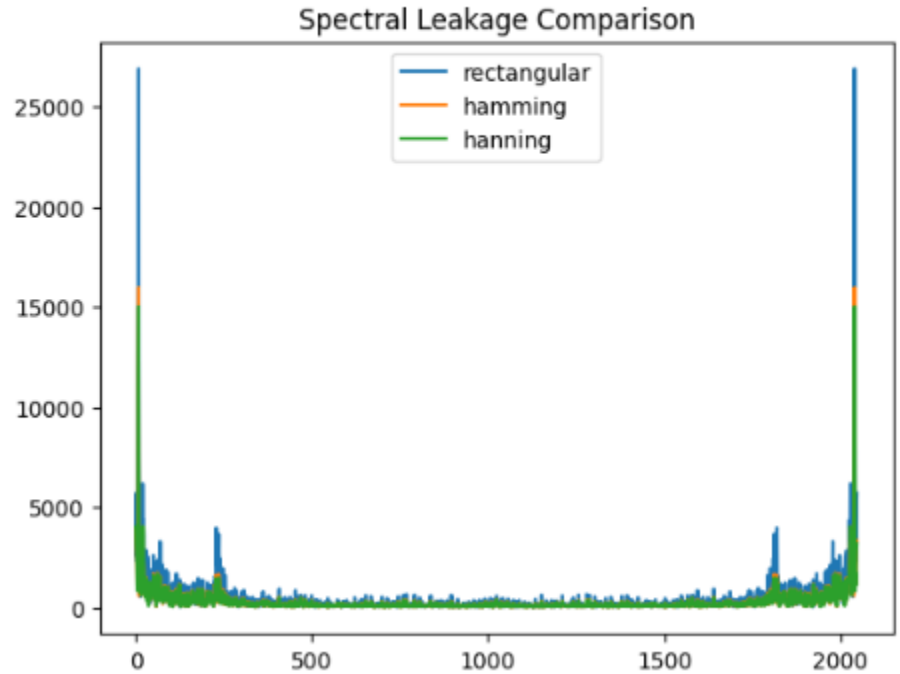

Figure 2. shows the spectral leakage behavior for different windowing functions. X-axis is the frequency bins (discrete frequencies after FFT). Y-axis amplitude (or power) of frequency components

Leakage metric: The computed leakage values for all window functions are very close (~0.97), indicating that the chosen leakage metric does not strongly differentiate between window types. However, visual inspection of the spectrum shows that Hanning and Hamming windows reduce side lobes compared to the rectangular window.

SNR Analysis: The rectangular window shows an infinite SNR due to zero estimated noise in the calculation, which is misleading. In practice, it exhibits the highest spectral leakage. Hamming and Hanning windows show finite SNR values (4.63 dB and 3.90 dB), indicating better suppression of spectral artifacts.

a. Voiced/Unvoiced Detection

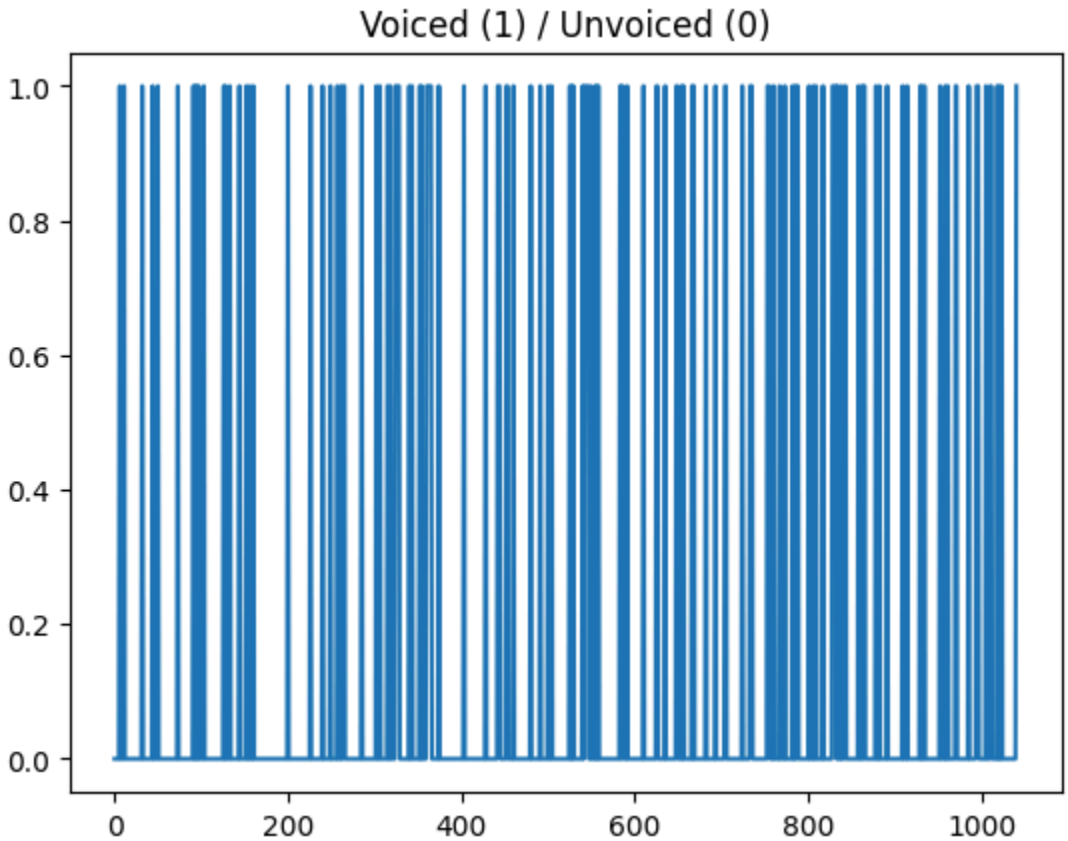

Figure3. X-axis is the frames(10msec step) and Y-axis is voiced (1) / unvoiced label(0).

A frame is classified as voiced if :-

- Energy > threshold
- ZCR < threshold
- Cepstral peak present

Visual inspection:-

- Voiced regions appear as continuous segments (value = 1)
- Unvoiced regions appear as gaps (value = 0)
- Raw predictions are noisy with frequent transitions
- Smoothing improves temporal consistency
- Boundary detection depends heavily on threshold selection

b. Boundary detection

The initial boundary detection produced a large number of closely spaced transitions due to noisy frame-level classification. By applying a minimum gap constraint of 50 ms, spurious boundaries were removed, resulting in a more realistic segmentation of the speech signal. Boundaries were identified at points where the classification changes:

- 0->1 transition -> start of voiced region
- 1->0 end of voiced region

Most phonemes last >= 50-100msec and anything shorted is likely noise or fluctuation.

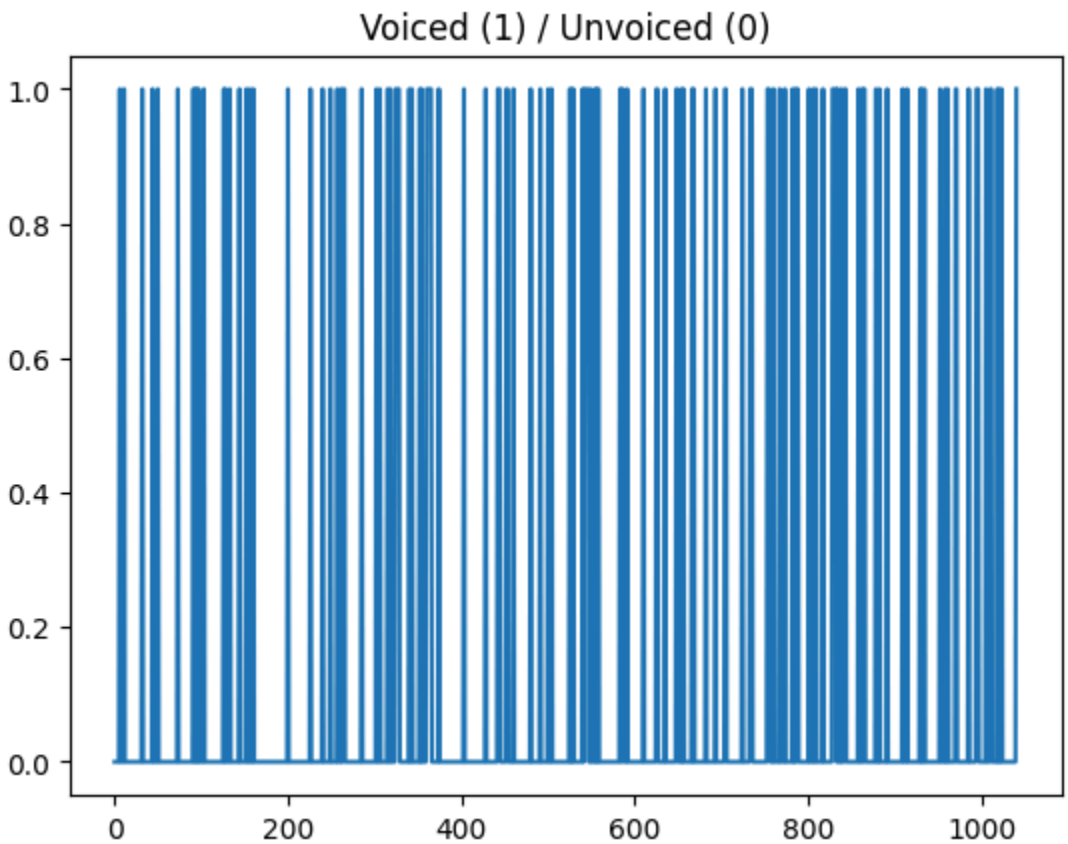

Figure 3: Smoothed voiced/unvoiced classification after temporal filtering.

c. Phonetics mapping

Phonetic mapping was performed using a pretrained Wav2Vec2 model. The model outputs token-level predictions over time, and phoneme boundaries were approximated by detecting changes in predicted tokens. These transitions correspond to changes in

acoustic patterns and serve as an estimate of phoneme boundaries. The goal is to Measure how close manual boundaries are to model boundaries

Steps used for manul boundary vs model boundary ➖

1. For each manual boundary find the closest model boundaries (ti_manual -tj_model)
2. Compute error
3. Appy RSME on error
4. Normalize RSME

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Transcription: HE HOPED THERE WOULD BE STEW FOR DINNER TURNIPS AND CARROTS AND BRUISED POTATOES AND FAT MUTTON PIECES TO BE LADLED OUT IN T
Manual boundaries: [3.61, 3.66, 5.42, 5.48, 8.3]
Model boundaries: [0.5608061420345489, 0.5808349328214971, 0.6008637236084453, 0.6409213051823417, 0.7010076775431863, 0.7210364683301345,
RMSE: 4.9631498481552105
Normalized RMSE: 0.47562528492143846
```

Results:-

- RMSE ≈ 4.44 sec
- Normalized RMSE ≈ 0.43

The high RMSE is due to:

- Mismatch in boundary density
- Different definitions of boundaries
- Temporal misalignment

Thus, RMSE alone is not a perfect metric for this task.

Conclusion

This work demonstrates the implementation of cepstral feature extraction and its application to speech segmentation. While the manual approach provides interpretability, it is sensitive to parameter tuning and noise.

In contrast, deep learning models provide more precise and robust segmentation due to learned representations. Future work can explore hybrid approaches or improved evaluation metrics for better alignment between methods.